

Optical Printing of Single Au Nanostars

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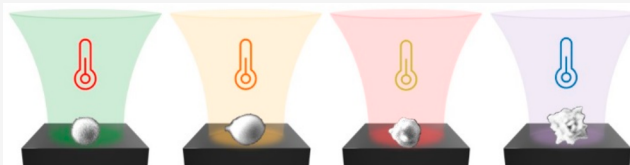
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Supporting Information

ABSTRACT: Obtaining arrays of single nanoparticles with three-dimensional complex shapes is still an open challenge. Current nanolithography methods do not allow for the preparation of nanoparticles with complex features like nanostars. In this work, we investigate the optical printing of gold nanostars of different sizes as a function of laser wavelength and power. We found that tuning the laser to the main resonances of the nanostars in the near-infrared makes it possible to avoid nanoparticles reshaping due to plasmonic heating, enabling their deposition at the single particle level and in ordered arrays.

KEYWORDS: optical lithography, optical forces, lasers, colloidal particles, single nanocrystals, plasmonic heating



Plasmonic nanoparticles (NPs) have outstanding optoelectronic properties due to their localized surface plasmon resonances (LSPR).¹ The frequency and strength of these resonances can be tuned with the NPs' size and morphology. Among the many available shapes, gold nanostars (AuNSs) stand out for their higher number of hot-spots and their stronger field enhancement near the tips.² For this reason, AuNSs have been extensively applied as substrates for surface enhanced Raman spectroscopy (SERS) and biosensing.^{3–7} Typically, they present LSPR close to the near-infrared window of biological tissues, making them suitable for photothermal therapies^{8–10} and cellular imaging.¹¹ Despite their irregular shape, they tend to present an isotropic interaction with light due to the many orientations of their tips, a property that was exploited to produce random lasing in a dye-doped gain medium.¹² Finally, the presence of optical hot-spots is also important in plasmon-assisted chemistry.^{13–15}

Deposition of AuNSs in planar surfaces is necessary for many of the aforementioned applications, and this has been achieved in most cases by drop-casting colloids, or through their electrostatic attachment to functionalized surfaces.^{16–18} Despite the widespread use of AuNSs, their deposition in ordered arrays remains a challenging task, and it is appealing for a systematic and robust correlation of the NP morphology with its performance. It is nearly impossible to create 3-D spiky Au NPs through the different available lithography methods,¹⁹ but a handful of works can be found in the creation of planar 2-D patterns or dimers.^{20,21} Only a few reports proved the successful preparation of arrays of AuNSs through growing spherical Au seeds previously organized by micellar self-assembly²² or by numerous steps of nanoimprinting lithography,^{23,24} lacking versatility in the positioning of the AuNSs. Although several methods based on self-assembly or directed-assembly have been presented to obtain arrays of complex-shaped nanocrystals over

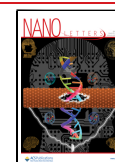
large areas,²⁵ none of them proved to be successful with AuNSs. One of the main constraints is the need for a monodisperse and homogeneous colloidal sample to be deposited, which is very difficult to achieve when having different lengths and numbers of spikes as in the case of AuNSs suspensions.

Optical printing is a method that allows the fabrication of arrays of single NPs employing colloidal suspensions and lasers, with the only requisite that the NPs absorb and/or scatter the light at the printing wavelength. This means that the NPs can have arbitrary shapes or compositions and that different kinds of NPs can be combined in the same substrate.^{26,27} Optical printing is based on the balance between the optical forces arising from NP illumination and the repulsive forces between the NP and the substrate.²⁸ Each type of NP has a particular optical response, so the optical forces depend on the wavelength and irradiance employed. One of the open challenges of this technique is the NP photostability when irradiated, especially in the case of anisotropic metallic NPs. Due to the typically high absorption cross section of metallic NPs, the photothermal heating during the printing process could lead to NP deformation, melting, or degradation of the surface functionalities.²⁷ The temperature rise depends on the absorption cross section and its dissipation to the surrounding media.²⁹ For example, an 80 nm Au nanosphere can reach more than 500 K during the printing process with a 532 nm laser at the lowest possible power. This temperature increase does not affect

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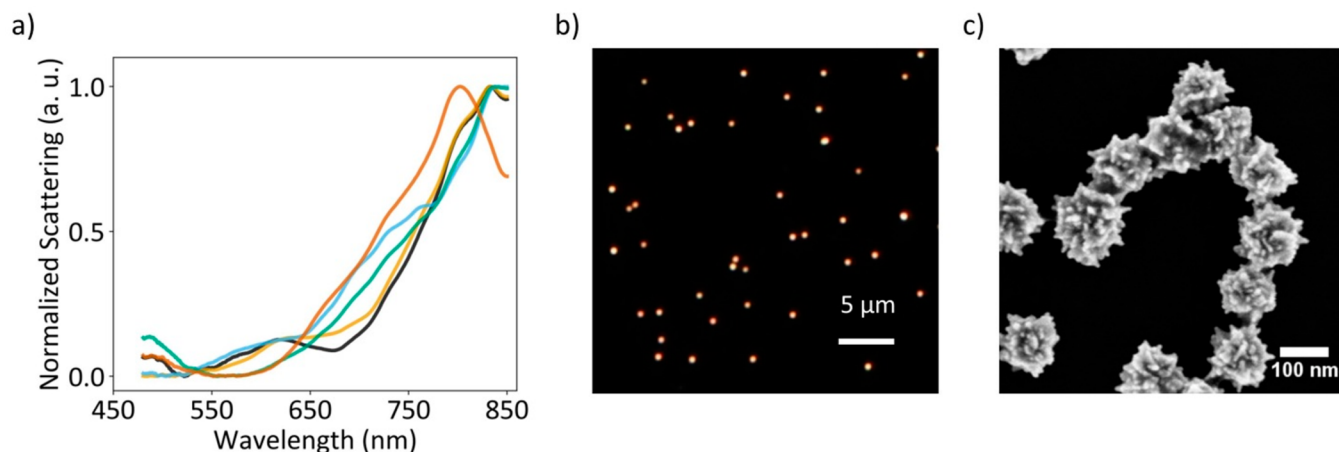


Figure 1. a) Normalized scattering of five single 130 nm AuNSs immersed on water and deposited onto glass by electrostatic attachment. b) Dark-field image of 130 nm AuNSs immersed on water and deposited onto glass by electrostatic attachment. c) SEM image of 130 nm AuNS deposited on a Si substrate by electrostatic attachment.

nanospheres shape, but it must be carefully considered when complex shapes and asymmetries are present, like in the case of nanorods, nanotriangles, and nanostars, because the mobility of atoms, driven by the surface energy minimization, is increased in the small sized tips and edges.^{30–34}

Therefore, the strategy to preserve the morphology of NPs during optical printing is to find the optimal printing wavelength that maximizes optical forces while minimizing the generated heat.²⁷ However, finding these conditions is not straightforward, especially for complex NPs for which the calculations of optical forces, heat generation, and dissipation are difficult or may involve a number of assumptions. In this work, we experimentally investigate the optical printing of AuNSs on glass substrates and assess different conditions aiming to produce arrays of intact AuNSs. We explore different printing wavelengths (532, 592, 642, and 808 nm) and irradiances. For each case, we quantify the produced damage to the NPs. In this way, we found that the optimal condition to preserve the morphology of AuNSs is to use an 808 nm laser tuned to their LSPR. On the contrary, using a 532 nm laser far from the LSPR reshapes the AuNSs into nanospheres during the printing process. Employing a nanothermometry technique with single particle resolution, we show that observed reshaping is indeed heat-induced. Finally, we demonstrate the resonant optical printing of single AuNS arrays from colloidal solutions of three different sizes (85, 115, and 130 nm in diameter), preserving their morphological and optical properties.

PVP-coated AuNSs were prepared by colloidal synthesis and then deposited onto glass substrates in two ways: by electrostatic attachment and by optical printing onto an electrostatically repulsive substrate. To this end, the glass substrates were functionalized with charged polymers (see [Supporting Information](#), SI). Electrostatic deposition was used as a control to characterize the unaltered AuNSs at the single particle level. For optical printing, linearly polarized 532, 592, 642, and 808 nm CW lasers focused to their diffraction limit were employed, each of them having a beam waist of 268, 310, 312, and 381 nm respectively (considered as Gaussian beams). More details about the setup can be found in our previous works,^{28,35} and in the SI.

Scattering spectra of five electrostatically attached single AuNSs are shown in [Figure 1a](#). The average scattering spectra (see [Figure S2](#) in SI) presents a maximum peaking at $\lambda_{\text{max}}^{\text{s}} = 813$

nm (standard deviation = 40 nm), resembling a bulk extinction spectrum of the AuNSs colloid.

A dark-field image of the sample is presented in [Figure 1b](#), showing that the AuNSs look uniform in color. Scanning electron microscopy (SEM) images of AuNSs deposited on a Si substrate are shown in [Figure 1c](#). A nominal diameter (d) of the AuNSs was calculated as $d = 2\sqrt{\frac{A}{\pi}}$, where A is the AuNS area obtained from the SEM image. This sample presented an average diameter of 129 nm (standard deviation = 9 nm, see histogram in [Figure S3](#) of the SI). For simplicity, this sample will be named “130 nm AuNSs” from now on.

Different arrays of 130 nm AuNSs were fabricated through optical printing employing the four different lasers mentioned before. Dark-field images of 10×10 AuNSs arrays separated by $3 \mu\text{m}$ in both directions are presented in [Figure 2a](#). It can be seen at first glance that the resulting arrays vary depending on the wavelength (λ) used. Printing at $\lambda = 532$ nm delivers the more imprecise and less uniform arrays, while the most homogeneous and regular arrays are obtained with $\lambda = 808$ nm. An intermediate behavior is observed for $\lambda = 592$ and $\lambda = 642$ nm lasers.

Each laser was set at the minimum power necessary to achieve the irradiance needed for printing, i.e., to overcome the repulsion barrier imposed for the substrate.²⁸ As the electrostatic barrier is independent of λ , the minimum optical force to print is the same in all cases. [Figure 2b](#) summarizes the minimum printing irradiance used for each λ (see SI for a detailed calculation of the irradiances). These values were compared with average scattering spectra. It could be seen that when λ is detuned from the plasmonic resonance higher irradiances are needed to achieve printing. As the optical force needed was the same, the lower interaction with the light had to be compensated by increasing the irradiance.

The outcome of each printing experiment was also characterized by scanning electron microscopy (SEM). In [Figure 2c](#), correlated SEM images and scattering spectra of single AuNSs printed with $\lambda = 532$, 642, and 808 nm are shown. Most of the AuNSs printed with $\lambda = 532$ nm suffer photothermal deformation, some transforming into spheres, others losing their tips and causing the scattering to blue shift with respect to the original AuNS (cases 2, 3). It was also found that the printing process could generate dimers of a spherical NP and a AuNS

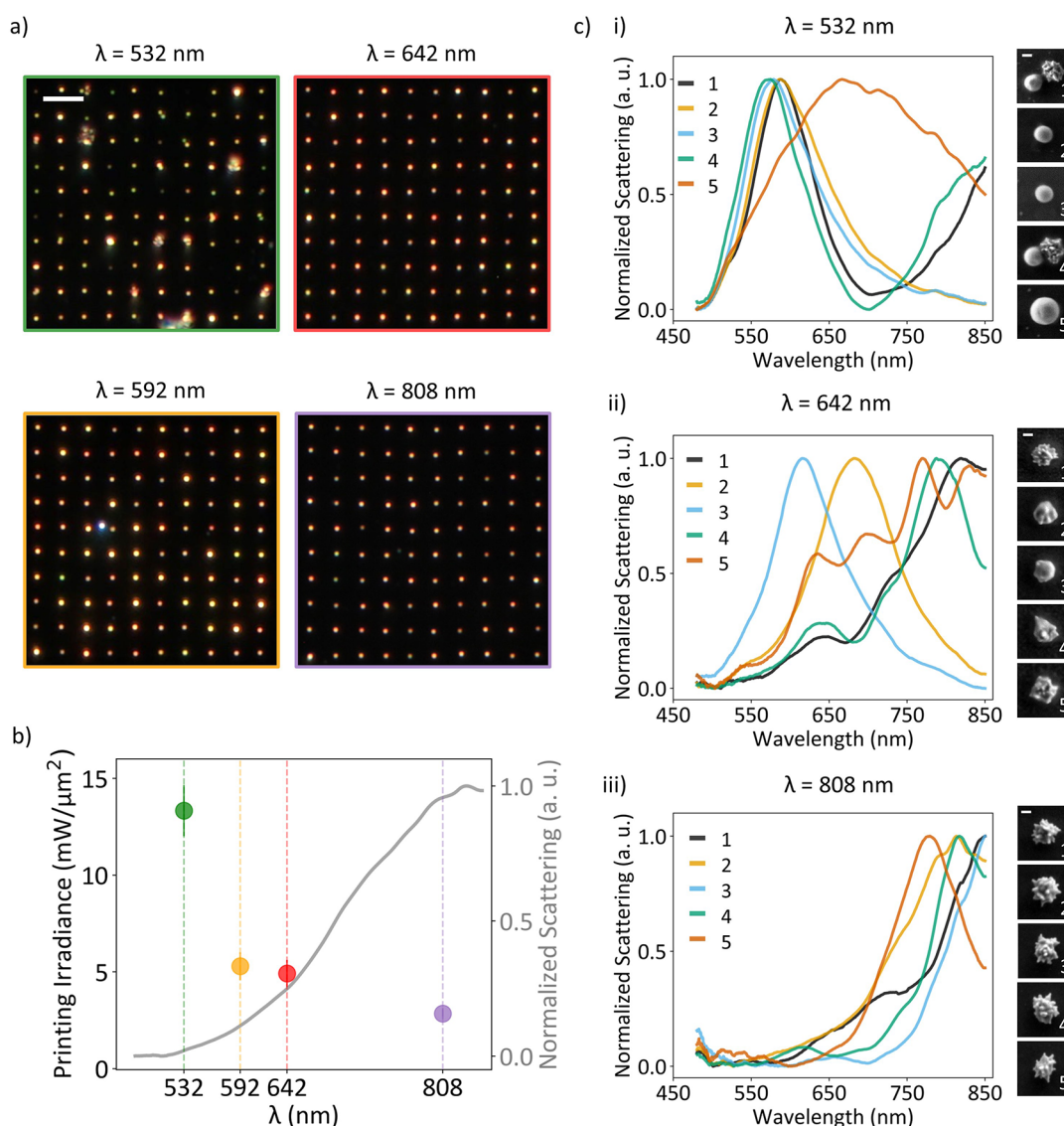


Figure 2. Optical printing AuNSs at different wavelengths. a) Dark field images of AuNSs arrays optically printed with different lasers: 532, 592, 642, and 808 nm. Scale bar = 5 μm. b) Printing irradiance vs laser wavelength λ (green, yellow, red, and violet dots). These irradiances are compared with the average scattering maximum of 39 single AuNSs measured after drop-casting the colloid (black solid line). c) Scattering spectra and SEM images of the different NPs printed at different λ : (i) 532 nm, (ii) 642 nm, and (iii) 808 nm. Scale bar in SEM images = 50 nm.

(cases 1, 4). Additionally, bigger spherical NPs were also observed (case 5), suggesting the fusion of a dimer to generate a bigger NP. At $\lambda = 642$ nm, the photothermal damage decreases significantly, with some AuNSs reshaped to spheres (case 3), while others conserved the scattering spectra and shape of the original AuNS (cases 1, 4). AuNSs that were not fully transformed into spherical NPs were also found, in the following called “semistars” (cases 2, 5). Finally, both the scattering spectra and SEM images show that AuNSs can be optically printed with preserved morphology using $\lambda = 808$ nm.

As it can be seen from the examples in Figure 2c, scattering spectra can be employed to rapidly assess whether a AuNS suffers photothermal deformation without the need of a *posteriori* SEM imaging. This approach was used to quantify the outcome of the printing process over 100 AuNSs printed at each λ . Scattering spectra were divided in three groups, according to their maximum ($\lambda_{\text{max}}^{\text{S}}$) and the full width at half-maximum (FWHM): stars, semistars, and spheres (more details can be found in Figure S4 of the SI). The results obtained for

NPs printed at 642 nm are presented in Figure 3a, where 10 spectra of each case are shown. SEM images were used to check this classification (Figure 3b).

Using this classification, the outcome of the printing process for each λ was quantified, as displayed in Figure 3c. It must be noted that, as shown in Figure 2c, 532 nm laser produces mainly spheres or aggregates. To simplify the analysis, the latter were included in the semistars group. This analysis confirms that 808 nm is the best laser wavelength to print the AuNSs while preserving their original properties, while 532 nm is the laser that produces more spheres or aggregates. The 642 and 592 nm lasers have a similar behavior, producing a partial reshaping of the AuNSs.

A series of experiments were carried out on electrostatically attached AuNSs on glass to gain a better understanding of the observed reshaping, which we ascribe to a plasmonic temperature rise. Although the absorption cross-section and dissipation of AuNSs onto the substrate are unknown parameters, the

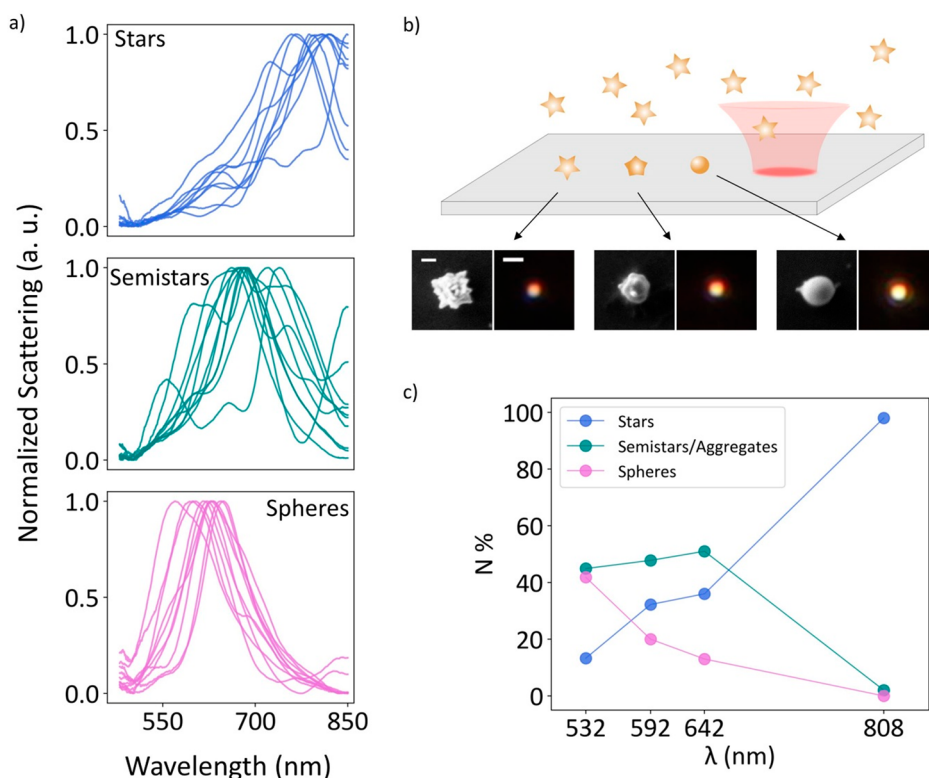


Figure 3. Classification of AuNSs optically printed at different wavelengths: stars, semistars, and spheres. a) Scattering spectra of 10 NPs per group obtained by printing at 642 nm. b) Scheme of the possible outcomes of optical printing (nanoparticles and laser beam are not-to-scale), along with a dark field and SEM image of a representative NP per group. Scale bar in dark field images = 1 μm . Scale bar in SEM images = 50 nm. c) Percentage of NPs type as a function of the laser wavelength. Each dot was obtained from a 10×10 array for each wavelength, printed at the minimum irradiance.

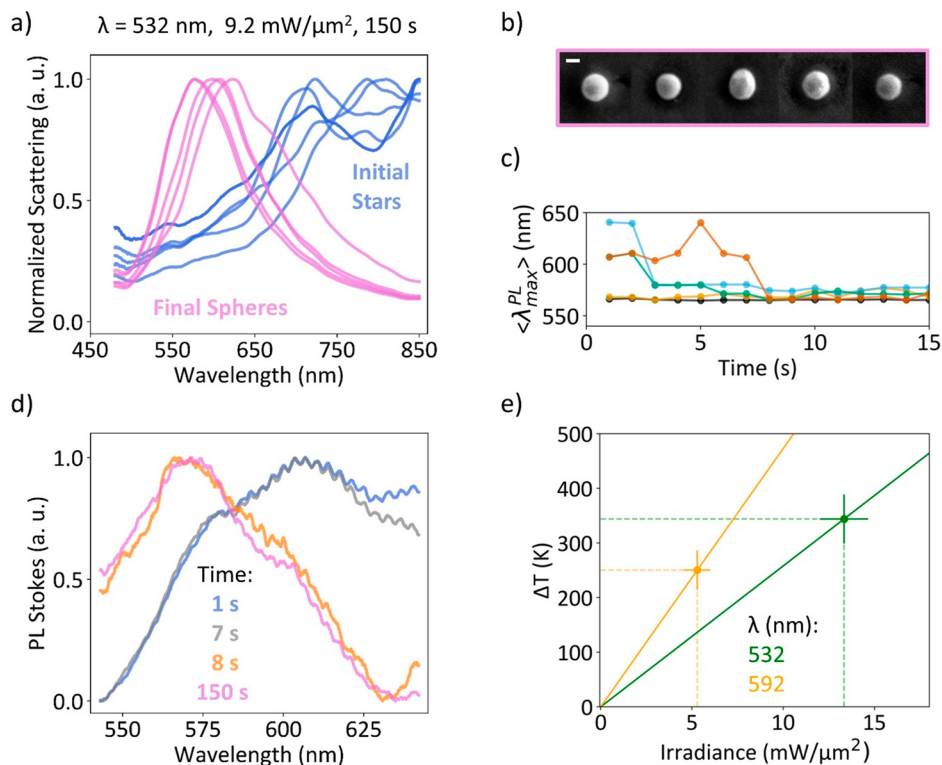


Figure 4. Reshaping of AuNSs by irradiation. a) Initial and final scattering spectra of five AuNSs after 150 s of irradiation at 532 nm, $9.2 \text{ mW}/\mu\text{m}^2$. b) Representative SEM images of five AuNS after the irradiation. Scale bar in SEM images = 50 nm. c) Maximum PL as a function of time for the five AuNSs. d) PL Stokes spectra of one AuNS measured at different times during the intentional reshaping at 532 nm. e) Temperature increment as a function of the irradiance for AuNSs irradiated at 532 and 592 nm. The dots account for the minimum printing irradiance at each wavelength.

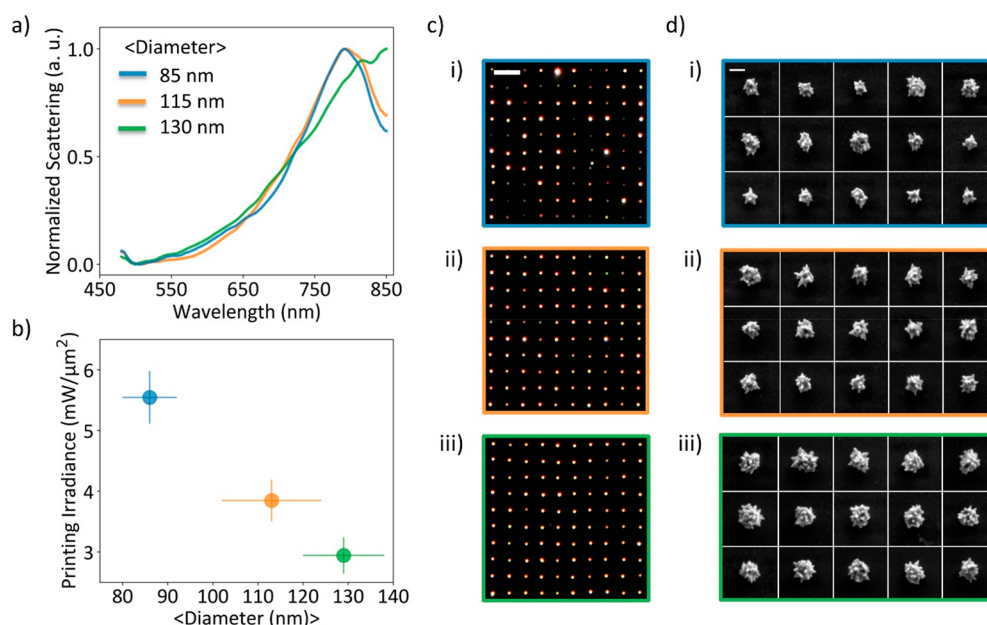


Figure 5. Different sized AuNSs optically printed with 808 nm. a) Normalized scattering average spectra of different sized AuNSs printed with 808 nm laser. b) Minimum printing irradiances vs mean diameters of the three sized AuNSs (each point color matches the label in a). c) Dark-field images of 10×10 arrays of the printed AuNSs (i) 85, (ii) 115, and (iii) 130 nm. Scale bar = $5 \mu\text{m}$. d) SEM images of 15 printed AuNSs: (i) 85, (ii) 115, and (iii) 130 nm. Scale bar = 100 nm.

printing outcome suggests that the temperature rise when using 532 nm is higher than with 808 nm.

Understanding the photothermal reshaping could be the key to improve the printing process of AuNSs or other NPs of complex shape. To study this, electrostatically deposited AuNSs were exposed to controlled irradiances during different periods of time. Figure 4a shows the scattering spectra of five AuNSs before and after irradiation with $\lambda = 532$ nm during 150 s with an irradiance $I = 9.2 \text{ mW} \cdot \mu\text{m}^{-2}$, which represents 70% of the minimum printing irradiance. In agreement with the observations during optical printing, the five AuNSs reshaped to spheres. SEM images of totally spherical NPs obtained after irradiation are displayed in Figure 4b (see more results in Figure S5). The reshaping process was studied in real time by measuring the Stokes photoluminescence (PL) emission spectra with a time resolution of 1 s. Figure 4c shows the maximum wavelength of PL vs time for the five AuNSs. Notably, in some cases, the reshaping occurs during the first second of observation, meaning that the time resolution is not enough to capture the fast-reshaping dynamics. In other cases, there is a delay until the onset of the reshaping, but once it starts it is completed within one second. Figure 4d shows an example of this last case, corresponding to the dark orange curve of Figure 4c. In there, the emission PL spectra of a single AuNS is shown at four different times, with the most significant changes occurring between 7 and 8 s.

In summary, preserving the morphology of AuNSs during optical printing with $\lambda = 532$ nm is not possible. However, similar experiments performed at $\lambda = 642$ nm showed a significantly slower reshaping that was not complete even after 150 s of illumination with irradiance values comparable to the ones used during printing (Figure S6).

To quantify the temperature rise of the AuNSs upon illumination, we applied hyperspectral nanothermometry based on anti-Stokes PL.²⁹ This method delivers the photothermal coefficient β which links the temperature increase to the

illumination irradiance as $\Delta T = \beta I$. The photothermal coefficient β was measured at 532 and 592 nm on several single AuNSs electrostatically attached onto glass substrate, surrounded by water, using irradiances that do not produce reshaping or bubbles ($I = 3.3 \text{ mW} \cdot \mu\text{m}^{-2}$ and $I = 2.2 \text{ mW} \cdot \mu\text{m}^{-2}$, respectively). The obtained results were $\beta_{532} = (26 \pm 2) \frac{\text{K}}{\text{mW} \cdot \mu\text{m}^{-2}}$ and $\beta_{592} = (47 \pm 2) \frac{\text{K}}{\text{mW} \cdot \mu\text{m}^{-2}}$. A brief account of the method, PL Stokes and anti-Stokes spectra, and histograms of β are presented in Figure S7 of the SI. ΔT vs I values calculated with these measured β are presented on Figure 4e. The dots mark the irradiance employed for optical printing at each wavelength. It could be observed that β_{592} is higher than β_{532} , but for the minimum printing irradiance, ΔT was higher at 532 nm than at 592 nm. An additional experiment of optical printing at 592 nm was performed employing a higher irradiance, so as to obtain a ΔT equivalent to the one reached while printing at 532 nm. The same output in terms of reshaping was obtained (Figure S8 of the SI), confirming that the reshaping is due to photothermal heating.

It is interesting to compare the aforementioned results with the previously observed heat-induced morphological change of AuNSs, shown by Vanrompay et al.³¹ In that work, the authors exposed the AuNSs to temperatures from 473 to 673 K, inside a transmission electron microscope, and studied the 3D shape of AuNSs as a function of time. For every temperature in the studied range, they observed redistribution of Au within the first minute. The amount of redistributed Au was larger at higher temperatures, losing almost all the AuNS tips during the first minute when heating at 673 K. These findings are in line with our own observations, in which comparable temperatures are reached during printing with a 532 nm laser. However, we observed a faster reshaping dynamic, requiring less than one second rather than several minutes. The different time scale could be explained by the larger thermal inertia when heating on the macroscale, and by the fact that the experiments were performed in vacuum instead of water.

Using the same colloidal synthesis method, by varying the ratio between the concentration of the gold precursor and the seeds (see experimental details in the SI), we prepared AuNSs with mean diameters of 85 and 115 nm (see histograms in Figure S9 of the SI). Optical printing of these two differently sized AuNSs leads to analogous results to those obtained for the 130 nm diameter AuNSs: 532 nm laser produces the maximum reshaping, while printing with 808 nm laser preserves the morphology and optical properties of the original AuNSs (see Figure S10 of the SI). When the nanostars reshape to spheres, the maxima of their scattering spectra follow the expected trend for the different sized final nanospheres (the smaller the NP diameter, the maximum is blue-shifted; see Figure S11 of the SI).

Figure 5a shows the normalized average scattering spectra of the three systems printed with 808 nm. The minimum printing irradiance at 808 nm as a function of the diameters is presented in Figure 5b. As can be seen, less irradiance was employed for larger particles, because the pressure radiation efficiency increases with the size, due to larger absorption and scattering. Finally, dark-field images of 10×10 arrays of each size are presented in Figure 5c, and in Figure 5d, SEM images of 15 printed AuNSs of each size are shown. In summary, the 808 nm laser can optically print all tested AuNSs sizes maintaining their original optical properties.

In conclusion, ordered arrays of spiky AuNSs of different sizes were obtained through optical printing. A careful optimization of the printing parameters was carried out to avoid reshaping of the AuNSs during the printing process. We found that when employing a laser with a wavelength close to the LSPR of the AuNSs, i.e., 808 nm, they can be printed preserving their shape and optical properties. On the contrary, when optical printing is performed with lasers detuned from the main plasmon resonance, reshaping of the AuNSs was observed. In the case of printing at 532 nm, a total spherization of the AuNSs takes place. At each wavelength, a different ratio between axial optical forces and photothermal heating is obtained. In resonance, this ratio is larger. This means that the optical forces needed to overcome the electrostatic barrier are achieved with less heating than in the nonresonant case. To prove this interpretation, confocal hyperspectral anti-Stokes thermometry was employed. It was found that the AuNSs reach ~ 650 K when printed with $\lambda = 532$ nm at the printing irradiance, while they reach ~ 550 K when printed at $\lambda = 592$ nm. It was also shown through an *in situ* measurement that reshaping to a spherical nanoparticle can occur in times of seconds for the higher temperatures. By contrast, at lower temperatures the reshaping is only partial even after 150 s of illumination.

In summary, this work establishes the best conditions to obtain arrays of AuNSs by optical printing and rationalizes the influence of the different parameters, paving the way for the optimization of optical printing of other nanoparticles with complex shapes.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.2c05109>.

Experimental methods, setup description, additional experiments and characterization (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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